7. Gregory, Paul R. and Stuart, Robert C. The Global Economy and Its Economic Systems, 2013 South-Western College Pub (2013).

Reference Books

1. Atkinson, R.L., Atkinson, R.C., Smith, E.E., Bem, D.J. and Nolen-Hoeksema, S. (2000). Hilgard's Introduction to Psychology, New York: Harcourt College Publishers.
2. Berne, Eric (1964). Games People Play – The Basic Hand Book of Transactional Analysis. New York: Ballantine Books.
3. Ferrell, O. C and Ferrell, John Fraedrich Business Ethics: Ethical Decision Making & Cases, Cengage Learning (2014).
4. Duane P. Schultz and Sydney Ellen Schultz, Theories of Personality, Cengage Learning, (2008).
5. Saleem Shaikh. Business Environment, Pearson (2007).
6. Chernilam, Francis International Buisness-Text and Cases, Prentice Hall (2013).
7. Salvatore, Dominick, Srivastav, Rakesh., Managerial Economics: Principles with Worldwide Applications, Oxford, 2012.
8. Peterson H. Craig. and. Lewis, W. Cris. Managerial Economics, Macmillan Pub Co; (1990).

Evaluation Scheme

S. No.	**Evaluation Elements**	**Weightage (%)**
1	MST	25
2	EST	45
3	Sessionals (Include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30

UME606 Machine Design-II

L	T	P	Cr
3	2	0	4.0

Course Objective: The objective for this course is to impart the knowledge required to apply the design procedure to the mechanical system design. The course is also intended to provide the experiential learning experience towards the design of commonly used mechanical systems.

Syllabus

Contents:
The focus of this course will be on the design of mechanical systems/subsystems based on the practical examples, such as the design of transmission and braking system of an automobile, power tool, and power operated machine tools etc. Students will be given exposure to various case studies related to the real-world mechanical systems from the mechanical engineering laboratories, and/or workshop and/or experiential learning center to understand the design philosophy and the design concept of the existing mechanical system/subsystem. Additionally, the design principles of following standalone machine elements which are not included in Machine Design-I will be discussed in detail.

- Sliding Contact Bearings
- Design and analysis of various types of Gears
- Design and analysis of Springs
- Design and analysis of Brake and Clutch mechanisms

The different tasks of the experiential learning activities may include hands-on experiences on disassembly and assembly of available mechanical systems e.g. engine and gear box, dimensioning, material recognition, design calculations, failure analysis, in addition to the consideration of manufacturing aspects such as fits, tolerances, surface finish, etc.

NB: Prescribed design data book will be allowed in MST and EST. ASTM or equivalent standard will be followed in the design of machine elements.

Design Assignment:
The design assignment will be based on the ongoing experiential learning activities and/or on the similar mechanical systems available in the mechanical engineering laboratory or workshop to understand the design approach of an integrated system comprising two or more of the machine elements. This assignment will essentially include hands-on experience in disassembly and assembly of available mechanical systems, the details on material selection, force analysis, designing of components on the basis of stress analysis, use of CAD/CAE tools and generation of production drawings. The re-design of the existing mechanical system will be a desirable component of this assignment. This assignment will also include technical report writing and seminar presentation.

Course Learning Objectives (CLO)
The students will be able to: